

Kushal Singh

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Education

Chandigarh University, Punjab, India
B.E. in Computer Science
CGPA: 7.94

2022 – 2026

Experience

Indian Institute of Technology (IIT BHU)

Jun 2025 – Jul 2025

Research Intern

- Conducted IoT traffic classification using bit-level data, achieving **92% accuracy** across device categories.
- Built lightweight Random Forest & CNN models using **128–256 bit inputs**, improving inference speed by **35%**.
- Designed efficient traffic analysis methods, reducing computational overhead by **40%** for low-power IoT systems.

WhatBytes

Mar 2025 – Apr 2025

Developer Intern

- Built Python & Django REST APIs with JWT, handling **1.5k+ requests/day** with **200ms** average latency.
- Integrated payment gateway and email-based auth, reducing login failures by **25%** and supporting **500+ users**.
- Automated tasks using Celery and created scraping pipelines, processing **10k+ records/day** and cutting manual effort by **40%**.

Vikas Parishad NGO

May 2024 – Jul 2024

Frontend Web Developer Trainee

- Developed responsive UI using HTML, CSS, JavaScript & Bootstrap, improving mobile responsiveness across **15+ pages**.
- Optimized website performance and accessibility, reducing load time by **30%** and improving Lighthouse score to **85+**.

Projects

EcoTrace – AI Powered Sustainability Analyzer (Live)

Python, Django, PostgreSQL, Gemini AI

- Built full-stack Django app analyzing environmental impact of **500+ products**, achieving **90% prediction accuracy**.
- Integrated Gemini AI API to generate CO₂ emissions and sustainability insights, reducing analysis time by **60%**.

PCAP Telecom Analyzer (GitHub)

Python, PyShark, Scapy, Pandas

- Built network traffic analyzer processing **1M+ packets** from PCAP files, extracting **10+ telecom metrics**.
- Parsed packets and generated structured CSV outputs (**50k+ rows**), reducing manual analysis time by **70%**.

NeuroSense AI – NICU Incubation Ward Monitoring System (GitHub)

Python, TensorFlow, Firebase, ESP32, JavaScript

- Developed real-time dashboard monitoring **5+ incubators** with live temperature & humidity streaming at **1-sec intervals**.
- Built LSTM model predicting temperature risks with **88% accuracy**, triggering smart alerts and reducing manual monitoring effort by **50%**.

Technical Skills

- **Programming Languages and Fundamentals:** C/C++, Python, JavaScript, OOPs, DSA
- **Cloud & DevOps:** Jupyter, Amazon Web Services, Git, Vercel, Docker
- **Database:** MySQL, NoSQL, PostgreSQL, MongoDB
- **Development:** Flask, FastAPI, REST API
- **Libraries & Frameworks:** Pandas, Matplotlib, NumPy, OpenCV
- **AI/ML:** Scikit-Learn, spaCy, TensorFlow, Py-Torch, CNN, RNN, Hugging Face, BERT
- **System Design:** Distributed Systems, Architectural Patterns

Publications

- Integrating Gene Expression and Imaging Data for Improved Cancer Diagnosis – IJEDR
- Unveiling the Power of Personalization in Health and Fitness Apps – IJRASET